

Based on 48 sources

Derivation Source/Method	Combinatorial/Algebraic Value	Physical Emergence Mode	Foundational Mathematical Principles	Role of Entropy/Information Torsion
Combinatorial Associator Path Count in Octonion Algebra $\sum$	137	Photon emergence as a quantized gradient from the contradiction field	Transition from non-associative 8D manifolds to real/complex hierarchies via 'Logic Tax'	Shannon and I Entropy maint symmetry and stability as cau
Octonion Algebraic Associator Mapping (HALIRA Framework)	137	Photon emergence as quantized gradient from contradiction field	Non-associative 8D manifolds (Octonions O), Contrarecursive Filtration, and Spectral Sequence Logic	Recursive Ent functions as a causal force; I Torsion acts as signal identify 'meaning resid
Algebraic Ladder transition from Quaternions to Octonions (Cayley-Dickson doubling)	Octonion path count 137 (α -fine equivalent)	Gravity and spacetime curvature emerging from the failure of perfect associativity	Hurwitz's theorem (Octonions as the largest normed division algebra); maximal non-commutativity with non-associativity	Osmotic press entropic gradi resolution; ass violations crea contradiction t
Logic Tax / Algebraic	137 (Path Count)	Transition from real to complex	Hegelian Algebraic Loop	Shannon and I Entropy maint

